Frontend - Theory Assignment 1

Q1. Explain the HTTP Request–Response Cycle with a suitable diagram. Describe the major steps involved when a client (browser) requests a resource from a web server and receives a response.

Answer: Whenever we visit a website, our browser and the web server talk to each other using a simple back-and-forth process called the HTTP Request-Response Cycle. The browser first asks (requests) for a webpage, and the server replies (responds) with the content of that page. This exchange happens every time we open a link or type a URL.

Here is how it works step by step:

1. The user types a web address or clicks on a link.
2. The browser prepares an HTTP request for the page or file that is needed.
3. This request travels over the internet to reach the correct web server.
4. The server reads the request and figures out what is being asked for.
5. The server sends back a response, which could be an HTML page, images, CSS styling, JavaScript, or other data.
6. The browser takes this response and renders it on the screen so the user can see the webpage.

Q2. Briefly explain the purpose and key features of HTML, CSS, JavaScript, Bootstrap, jQuery, and React.js.

HTML: HTML stands for HyperText Markup Language, and it is the backbone of every webpage. It defines the skeleton of a page using tags like headings, paragraphs, images, links, and forms, so the browser knows what content to display and how it is organized.

CSS: CSS, short for Cascading Style Sheets, is what makes a plain HTML page look attractive. It handles the visual side of things - colors, fonts, spacing between elements, borders, and the overall layout of the page.

JavaScript: JavaScript brings a webpage to life by making it interactive. It can react to a button click, check if a form is filled correctly, update content without reloading the page, and much more.

Bootstrap: Bootstrap is a ready-to-use frontend toolkit built with CSS and JavaScript. Instead of writing styling from scratch, developers can use Bootstrap's pre-built classes and components to quickly build websites that look good on any screen size.

jQuery: jQuery is a lightweight JavaScript library that simplifies common tasks such as selecting page elements, handling clicks and other events, and creating animations. It became popular because it let developers do more with fewer lines of code.

React.js: React.js is a JavaScript library used for building interactive user interfaces, especially single-page applications. It lets developers break a page into small, reusable components and update only the parts of the page that actually change, which makes apps faster and easier to manage.

Q3. What are client-side technologies and server-side technologies? Explain the difference between them with suitable examples.

Answer: Client-side technologies are the ones that work directly inside the user's browser. They take care of everything the user sees and interacts with on a webpage, such as the layout, colors, and button clicks. Common examples include HTML, CSS, and JavaScript.

Server-side technologies, on the other hand, operate on the web server rather than the browser. Their job is to handle incoming requests, connect with databases, and prepare the information that eventually gets sent to the browser. Examples include Python with Flask, PHP, Java, and Node.js.

Point	**Client-Side**	**Server-Side**
Runs on	User's browser	Web server
Main use	Design and interaction	Processing and handling data
Examples	HTML, CSS, JavaScript	Python/Flask, PHP, Node.js

Q4. What are block-level elements in HTML? Explain their characteristics and provide at least three examples.

Answer: Block-level elements are HTML tags that automatically move to a new line and stretch across the full width available to them. Because of this behavior, they are commonly used to build the main structure and sections of a webpage.

Key characteristics:

- They always begin on a new line rather than staying next to other content.
- By default, they take up the entire width of their parent container.
- Multiple block-level elements can be stacked one after another.
- They are ideal for splitting a page into clear, organized sections.

Common examples: <div>, <p>, <h1>, <section>, and <header>.

Q5. What are semantic elements in HTML? Explain their importance in web development and provide suitable examples.

Answer: Semantic elements are HTML tags whose names clearly indicate the purpose of the content they hold, instead of being generic containers. This makes the layout of a webpage much easier to understand just by looking at the code.

These elements matter because they keep the code organized and easier to maintain over time. On top of that, they help search engines index the page correctly and allow screen readers and other accessibility tools to interpret the content properly for users who need them.

Common examples: <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.

Q6. Write HTML code to create the table with Sr. No., Name, Designation, and Department.

```html
<table border="1">

  <tr>

    <th>Sr. No.</th>

    <th>Name</th>

    <th>Designation</th>

    <th>Department</th>

  </tr>

  <tr>

    <td>1</td>

    <td>John</td>

    <td>Manager</td>

    <td>Sales</td>

  </tr>

  <tr>

    <td>2</td>

    <td>Smith</td>

    <td>Sr. Manager</td>

    <td>Marketing</td>

  </tr>

  <tr>

    <td>3</td>

    <td>Jane</td>

    <td>Manager</td>

    <td>HR</td>

  </tr>

</table>
```

Q7. Write HTML code to create the nested list using an appropriate combination of ordered/unordered lists.

```html
<ul>

  <li>River

    <ul>

      <li>Ganga</li>

      <li>Yamuna</li>

      <li>Brahmaputra</li>

      <li>Narmada</li>

    </ul>

  </li>

</ul>
```

Q8. Explain the different types of CSS used for applying styles to an HTML document. Describe each type with a suitable example.

Answer: CSS can be added to a webpage in three main ways:

1. Inline CSS

This method adds styling straight into an HTML tag using the style attribute. It works best when you need to style just one single element.

```html
<p style="color: blue;">Hello</p>
```

2. Internal CSS

Here, the CSS rules are placed inside a <style> tag within the <head> section of the HTML file. It suits situations where only one page needs its own unique styling.

```html
<style>

  p { color: blue; }

</style>
```

3. External CSS

This approach keeps all the CSS rules in a separate file (with a .css extension) that gets linked to the HTML page. It is the most practical option when the same styles need to apply across many pages.

```html
<link rel="stylesheet" href="style.css">
```

Q9. Differentiate between HTML, CSS, and JavaScript based on their role in developing a web page. Explain what each technology contributes to a website.

Answer: HTML, CSS, and JavaScript each play a distinct role, and together they form a fully working webpage.

Technology	Role	What it adds
HTML	Structure	Headings, paragraphs, images, links, forms, etc.
CSS	Design	Colors, fonts, spacing, layout, and responsiveness
JavaScript	Interaction	Actions, events, dynamic content, and form validation

Q10. Answer the following short questions.

a. What is the role of Bootstrap in web development?

Answer: Bootstrap offers a collection of pre-built CSS classes and ready-made components, which saves developers time and effort. It makes it much easier to build websites that look polished and adjust well to different screen sizes.

b. What is jQuery and why was it widely used in web development?

Answer: jQuery is a JavaScript library designed to simplify everyday coding tasks. Developers relied on it heavily in the past because it made things like selecting elements, handling events, and creating animations possible with far less code than plain JavaScript.

c. What is React.js and what problem does it solve in frontend development?

Answer: React.js is a JavaScript library that helps developers build user interfaces using small, reusable pieces called components. It solves the problem of efficiently updating and managing parts of a webpage that change often, without having to reload the whole page.